

Nitya Patel

+91-9131174431 | nityapatel19@gmail.com | [linkedin.com/in/nityapatel19](https://www.linkedin.com/in/nityapatel19) | github.com/nityapatel19 | nityapatel19.com

EDUCATION

IIIT Vadodara

B.Tech in Computer Science | CPI: 8.61/10.0

Gandhinagar, Gujarat

2020 – 2024

EXPERIENCE

Graviton Research Capital

Software Developer – Tech Tools

Gandhinagar / Gurugram

June 2024 – Present

- Develop and maintain daily automations and exchange-specific workflows in Python and Bash.
- Built and own the `config_converter` module, which automates input expansion and reduces manual configuration effort.
- Designed an inventory allocation algorithm based on request patterns and usage history, preventing resource exploitation by any single candidate.
- Create internal tooling to eliminate repetitive tasks and improve team productivity.

VizExperts

Software Development Intern

Gurugram

January 2024 – May 2024

- Assisted with bug resolution in an AR/VR application built on Unreal Engine using C++ and Blueprint scripting.
- Designed and built an AI-based live drift correction module from scratch using Python, OpenCV, and PyTorch, targeting iPad AR workflows without multi-camera hardware support.
- Achieved a $\sim 10\text{--}15\times$ reduction in drift accumulation rate compared to baseline.

MathWorks

EDG Intern

Bangalore

May 2023 – July 2023

- Collaborated with the team on enhancements of the FMU3 Import Block.
- Shadowed experienced engineers to gain insights into providing technical support.
- Participated in various team events including a hackathon.

iHub-Data, IIIT Hyderabad

Machine Learning Intern

Hyderabad

May 2022 – July 2022

- Contributed to a cutting-edge Dataset Foundation project to create a platform for ML/AI researchers.
- Collaborated with a team of 10 to design the project architecture and establish an efficient delivery pipeline.
- Engineered a high-performing, multithreaded download manager for datasets up to 100 GBs.

PROJECTS

Ornithological Sound Identifier | *Python, PyTorch*

- Implementing a multilayered neural network that identifies birds from soundscapes.

Focused | *ReactJS, Django Rest Framework*

- Developing a virtual study/work platform enabling users to study/work together with minimal distractions.

Todoist Command Line Utility | *Python*

- Developed a command line utility for Todoist using the Todoist REST API.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, Matlab, SQL, Bash

Tools & Frameworks: Git, Django, ReactJS, discord.py, $\text{\LaTeX}$, Firebase, PyTorch

Operating Systems: Windows, Linux

ACHIEVEMENTS

Giftabled Hackathon: 2nd rank among 100 participants

CTF: 3rd rank among 100 participants

Tech Hunt: 1st rank among 150 participants